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A dual-stage fourier neural operator for Lamb wave scattering with physics embedded and residual refinement

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ABSTRACT

Lamb wave scattering provides a basis for extracting meaningful information from wave propagation in plates and offers essential theoretical support for nondestructive evaluation and structural health monitoring. However, scattering analysis remains challenging due to inherent dispersion characteristics and high sensitivity to geometric variations. Conventional numerical methods become computationally expensive for large spatial or high frequency domains. Physics informed neural networks sometimes suffer from slow convergence and unstable optimization when modeling high frequency or strongly heterogeneous wavefields. Fourier neural operators (FNOs) may suppress high frequency components that are essential for scattering analysis. To address these limitations, this paper proposes a dual-stage FNO framework with physics embedded and residual refinement for Lamb wave scattering modeling. The physics embedded stage establishes a globally consistent mapping between thickness distributions and scattering wavefields to capture the dominant propagation behavior, while the residual refinement stage learns discrepancies between predicted and reference wavefields to enhance the representation of scattering characteristics. The proposed framework is validated through numerical simulations with random thickness defects and laser scanning vibration measurement experiments. Results demonstrate that the proposed approach improves global prediction accuracy and physical consistency while yielding pronounced error reduction in defect regions, demonstrating its effectiveness for Lamb wave scattering analysis.

1. Introduction

Guided wave scattering refers to the alteration of wave propagation caused by discontinuities such as geometric irregularities, thickness variations, or material inhomogeneities, ultimately modifying the propagation path, energy distribution, and dispersion characteristics. It has long attracted widespread attention due to the inherent involvement of boundary conditions and mode conversion in scattering. Lamb waves represent the most typical modes propagating in plate-like structures. Although classical, Lamb wave scattering remains the cornerstone for understanding more complex situations involving irregular scatterers, multimodal interactions and intricate boundary conditions. Accurate modeling and interpretation of Lamb wave scattering not only provide crucial insights into wave interactions but also play a vital role in advancing nondestructive testing (NDT) and structural health monitoring (SHM) technologies for ensuring the integrity and reliability of engineering structures.

In recent decades, extensive analytical efforts have been devoted to understanding guided wave scattering in plate structures. Achenbach [1] [2] [3] [4] developed a unified theoretical framework for elastic wave dynamics in plates based on the elastodynamic reciprocity theorem. By establishing modal orthogonality and employing bounded virtual solutions to determine expansion coefficients, the derived formulation enabled an elegant and complete reconstruction of wave responses generated by point sources at arbitrary locations and orientations. This theoretical system avoids the intricate integral transforms commonly seen in classical formulations and has laid a solid foundation for subsequent advances in plate wave analysis and ultrasonic guided wave NDT. Building upon this foundation, Grahn [5] and Cegla et al. [6] systematically investigated the scattering of fundamental Lamb modes by asymmetric circular blind holes. Both studies adopted the wave function expansion method, expressing the interior and exterior wavefields as complete sets of Lamb and SH modes and solving for the unknown expansion coefficients through the enforcement of stress, displacement, bending moment, and shear force boundary conditions. Their work provided accurate predictions of mode conversion in

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scattering and demonstrated a clear route of theoretical evolution. Grahn [5] formulated a full three dimensional (3D) elastodynamic model complemented by Poisson-Kirchhoff plate theory for low frequency validation, whereas Cegla [6] improved the combined model by incorporating Poisson-Mindlin theory, significantly enhancing the prediction accuracy of scattering in high frequency regime and experimentally validating the S0 to A0 conversion predictions. Further generalization was achieved by Moreau et al. [7] [8], who extended the projection integral framework using Fourier series decomposition to address angular dependencies of non-circular cavities. The semi-analytical model provided a unified solution applicable to arbitrary through-thickness cavity shapes, with good agreement observed against finite element benchmarks in predicting multimodal scattering and energy redistribution. A major methodological progression came from the Complex Mode Expansion and Projection (CMEP) method introduced by Poddar and Giurgiutiu [9]. While retaining the essential philosophy of modal expansion, CMEP innovatively introduced an energy flow based vector projection strategy, where stress and displacement boundary conditions are projected onto conjugate modal spaces and solved within a power balance framework. This formulation not only imposes power conservation naturally but also improves numerical stability and convergence. The versatility of CMEP has been demonstrated across a wide range of applications, from thickness steps and welds [10] [11], to horizontal cracks and disbonds [9], and further to stiffened plates containing root cracks [12]. These studies underscore the strength of frameworks based on modal expansion in addressing guided wave scattering problems. Despite their rigor and physical transparency, these analytical and semi-analytical approaches rely on precise mathematical descriptions of governing equations and boundary conditions, involving extensive derivations and large matrix systems. More importantly, their effectiveness fundamentally depends on the ability to parameterize defect geometries in a manner compatible with modal expansions. Therefore, current analytical and semi-analytical frameworks are primarily limited to regular or parameterizable scatterers. Extending these powerful techniques to realistic defects with arbitrary geometric complexity, while easing the theoretical and computational burdens, remains an important challenge and a key direction for future research.

In parallel with analytical developments, numerical simulation has also become an indispensable tool for studying guided wave scattering. Despite differences in implementation, numerical methods fundamentally operate by discretizing the governing elastodynamic equations, which allows the originally continuous scattering problem to be recast as a finite dimensional system. Finite difference schemes employ structured grids to approximate spatial and temporal derivatives [13], whereas finite element methods discretize the waveguide into triangular or quadrilateral elements to accommodate complex geometries [14] [15]. Boundary element methods reduce the dimensionality by discretizing the structural boundary, yet still rely on numerical quadrature and dense boundary matrices [16]. Semi-analytical finite element formulations offer a more compact alternative by exploiting periodicity or geometric invariance, thereby further reducing the number of degrees of freedom [17]. Additional hybrid strategies [18] [19], spectral element approaches [20] and discontinuous Galerkin variants [21], continue this development by combining localized discretization with higher order basis functions or enhanced boundary treatments. These advances reflect a clear trajectory in numerical scattering analysis, where dimensionality reduction or discretization is used to reduce computational costs, and more advanced radiation conditions are adopted to better represent unbounded domains. While numerical methods provide the flexibility to accommodate arbitrary geometries and heterogeneous material distributions, they inherently face trade-offs among accuracy, stability, and computational efficiency. High frequency Lamb wave simulations demand extremely fine spatial discretization (often exceeding ten grid points per minimum wavelength), causing the computational burden to increase rapidly with excitation frequency, domain size, and defect complexity [22] [23]. Therefore, high fidelity scattering simulations remain computationally expensive, particularly in inversion or imaging tasks that require large parametric sweeps or repeated forward solves. These limitations motivate the exploration of data-driven and physics informed learning frameworks that aim to preserve the fidelity of numerical solvers while mitigating their computational bottlenecks.

In recent years, neural networks have emerged as powerful, nearly universal function approximators capable of addressing a wide range of engineering problems. Despite their impressive performance, concerns regarding interpretability and physical consistency remain. Since wave propagation is essentially governed by partial differential equations (PDEs), an important research attempts to directly embed physical laws into the learning process. Physics informed neural networks (PINNs) follow this paradigm by incorporating PDE residuals, boundary conditions, and initial conditions into the loss function, enabling the network to learn mappings that remain consistent with the underlying physics [24] [25] [26]. PINNs have since been applied to various wavefield problems, including acoustic wave simulation [22] [27], elastic wave inversion [28] [29], and ultrasonic imaging [30] [31], demonstrating their ability to recover continuous fields from sparse data and to alleviate the dependency on large training datasets. However, their reliance on high order automatic differentiation and complex multi-term residual losses often leads to slow convergence and unstable optimization, especially when modeling high frequency, dispersive, or strongly heterogeneous wavefields. Therefore, PINNs are typically well suited for solving specific modeling problems, but struggle to generalize across many input configurations or to efficiently approximate families of PDE solutions [32] [33].

This limitation has motivated the idea of neural operator learning, which seeks to approximate mappings between infinite dimensional function spaces rather than solutions to specific PDE instances. Representative frameworks include DeepONet [34] [35] [36] and Fourier neural operator (FNO) [37]. Unlike PINNs, which solve a particular PDE for a given input, neural operators are designed to learn a family of PDE solutions parameterized by varying coefficients, source terms, or boundary geometry. DeepONet employs a branch-trunk architecture that separately encodes functional inputs and spatial coordinates, making it conceptually flexible but computationally expensive when dense spatiotemporal sampling is required. Its pointwise evaluation mechanism also makes capturing long range wave propagation or multi-modal interactions more challenging. FNO, in contrast, performs global convolution in the spectral domain through fast Fourier transform (FFT), enabling efficient capture and generalization of the dependency relationships in wavefield propagation [38] [39] [40]. This strategy may be particularly attractive for guided wave problems, where wave propagate over large distances and accompany with dispersion, mode conversion, and scattering. The ability of FNO to model global interactions with relatively few spectral modes provides a clear advantage over architectures that rely solely on localized convolutions or pointwise sampling. However, the implementation of FNO involves truncating high frequency Fourier modes during spectral processing, which may suppress the high frequency and defect sensitive components crucial for accurate Lamb wave scattering analysis [41]. In addition, there are some surrogate models based on PINN and neural operators that have been successfully applied in different fields. However, some methods [42] [43] [44] mainly focus on improving the PDE consistency of global solutions, while others [45] [46] [47] aim to improve prediction performance through feature fusion or architecture coupling. These methods have not yet specifically solved the limitations caused by FNO spectrum truncation, nor have they considered the multi-scale characteristics of guided wave propagation and scattering. These considerations highlight the need to adapt and enhance neural operator frameworks when applying them to guided wave scattering problems. In particular, combining FNO with physical constraints and a residual learning strategy provides a pathway to retain its strengths in capturing global wave

behavior while compensating for its limitations in representing localized or high frequency scattering details. This motivates the dual-stage FNO framework developed in this study.

To address these challenges and extend the applicability of neural operator techniques to guided wave scattering, this study enhances the FNO framework by embedding the governing equation directly into the training process as physics constraints, followed by incorporating a residual refinement stage that models the discrepancy between predictions and reference wavefields. A finite difference scheme is employed to solve the Kirchhoff-Love plate equation and generate the out-of-plane displacement wavefields for dataset construction. The PDE embedding constrains the learned operator to remain consistent with physical laws and enhances its ability to represent global wave propagation behavior, while the residual refinement stage improves the representation of localized scattering characteristics that are often attenuated during spectral truncation in standard FNO architectures. The performance enhancement of the proposed dual-stage FNO is quantitatively evaluated by comparing wavefield predictions obtained with and without physics embedded and residual refinement. Additional validation is provided through laser scanning vibration measurements performed on an aluminum plate containing a randomly generated defect. Both numerical simulations and experimental observations demonstrate that the dual-stage FNO improves the accuracy and physical consistency of the predictions, thereby strengthening the operator learning framework for Lamb wave scattering. This approach also offers a promising direction for extending neural operator techniques to more complex structural configurations and practical nondestructive evaluation.

2. Methodology

2.1. Kirchhoff-Love plate theory

In an isotropic thin plate where the thickness is much smaller than its in-plane dimensions, the propagation of flexural (A0 mode) Lamb waves can be effectively described by the Kirchhoff-Love plate theory. This theory assumes that straight lines normal to the mid-surface before deformation remain straight and normal after deformation, and that transverse shear deformation is negligible [28]. Under these assumptions, the out-of-plane displacement $u_z(x,y,z,t)$ of the plate is identical to the mid-surface deflection $w(x,y,t)$. Therefore, the dynamic behavior of the plate can be fully characterized by $w(x,y,t)$. Based on the small deflection and linear elastic assumptions, the bending and twisting moments are related to the mid-surface curvature as follows [5]:

$$\begin{aligned} M_x &= -D\left(\frac{\partial^2 w}{\partial x^2} + \nu\frac{\partial^2 w}{\partial y^2}\right), \\ M_y &= -D\left(\frac{\partial^2 w}{\partial y^2} + \nu\frac{\partial^2 w}{\partial x^2}\right), \\ M_{xy} &= -D(1-\nu)\frac{\partial^2 w}{\partial x \partial y}, \end{aligned} \tag{1}$$

where the bending stiffness $D(x,y)$ depends on the local thickness $h(x,y)$ and is given by:

$$D(x,y) = \frac{Eh(x,y)^3}{12(1-\nu^2)}. \tag{2}$$

E and ν are Young's modulus and Poisson's ratio of the thin plate, respectively. The transverse shear forces per unit width are related to the spatial derivatives of the bending and twisting moments:

$$Q_x = \frac{\partial M_x}{\partial x} + \frac{\partial M_{xy}}{\partial y}, Q_y = \frac{\partial M_y}{\partial y} + \frac{\partial M_{xy}}{\partial x}. \tag{3}$$

The equilibrium of an infinitesimal plate element in the transverse direction requires:

$$\frac{\partial Q_x}{\partial x} + \frac{\partial Q_y}{\partial y} + q = \rho h(x,y)\frac{\partial^2 w}{\partial t^2}, \tag{4}$$

where $q(x,y,t)$ is the distributed transverse load per unit area, ρ is the material density and t is the time of propagation. Substituting Eq. (3) into Eq. (4) yields the dynamic equilibrium written in terms of bending moments:

$$\frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} + q = \rho h\frac{\partial^2 w}{\partial t^2}. \tag{5}$$

Finally, substituting Eq. (1) into Eq. (5) gives the governing PDE for the transverse deflection $w(x,y,t)$:

$$\nabla^2\left(D(x,y)\nabla^2 w(x,y,t)\right) + q(x,y,t) = \rho h(x,y)\frac{\partial^2 w(x,y,t)}{\partial t^2}, \tag{6}$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is the Laplace operator. For plates with uniform thickness, $D(x,y)$ is constant and Eq. (6) reduces to the classical biharmonic form:

$$D\nabla^4 w + q = \rho h\frac{\partial^2 w}{\partial t^2}. \tag{7}$$

2.2. Finite difference discretization of Kirchhoff-Love equation

To numerically compute the flexural wavefield governed by Kirchhoff-Love equation, a finite difference discretization is employed on a uniform Cartesian grid. Let the solution domain be divided into grids N_x, N_y and N_t respectively in the spatial and temporal dimensions. Then the domain is discretized as $x_i = i\Delta x$ ($i = 0,1,\ldots,N_x-1$), $y_j = j\Delta y$ ($j = 0,1,\ldots,N_y-1$), and $t_n = n\Delta t$ ($n = 0,1,\ldots,N_t-1$), where Δx, Δy, and Δt denote the spatial and temporal increments, respectively. The second-order spatial derivatives in the governing equation are approximated using the central finite difference:

$$\left.\frac{\partial^2 w}{\partial x^2}\right|_{i,j} \approx \frac{w_{i+1,j} - 2w_{i,j} + w_{i-1,j}}{(\Delta x^2)}, \left.\frac{\partial^2 w}{\partial y^2}\right|_{i,j} \approx \frac{w_{i,j+1} - 2w_{i,j} + w_{i,j-1}}{(\Delta y^2)}, \tag{8}$$

and

$$\left.\frac{\partial^2 w}{\partial x \partial y}\right|_{i,j} \approx \frac{w_{i+1,j+1} - w_{i+1,j-1} - w_{i-1,j+1} + w_{i-1,j-1}}{4\Delta x \Delta y}. \tag{9}$$

Based on these discrete spatial derivatives, the bending and twisting moments at each grid node are evaluated according to Eq. (1). The divergence of the moment field then provides the internal restoring force per unit area:

$$f\left[w_{i,j}\right] = \frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2}. \tag{10}$$

Substituting this discrete operator into Kirchhoff-Love equation yields the following explicit time-marching scheme:

$$w_{i,j}^{n+1} = 2w_{i,j}^n - w_{i,j}^{n-1} + (\Delta t)^2\left(\frac{f\left[w_{i,j}^n\right]}{\rho h_{i,j}} + \frac{q_{i,j}^n}{\rho h_{i,j}}\right). \tag{11}$$

The superscript n represents the $n-th$ time step. Equation (11) governs the temporal evolution of the mid-surface deflection and serves as the numerical basis for modeling the propagation of A0 mode Lamb waves in the thin plate. In program design, this process is implemented through 3×3 convolution kernels. Absorbing boundary conditions are applied to suppress unwanted edge reflections, while the stability of the explicit scheme is ensured by selecting a sufficiently small time step that satisfies the Courant-Friedrichs-Lewy (CFL) criterion.

2.3. Fourier neural operator

To construct a data-driven surrogate model capable of mapping thickness to its corresponding wavefield, a FNO network was developed. The network aims to approximate the nonlinear operator between the spatial thickness distribution $h(x,y)$ and the out-of-plane displacement field $\widetilde{w}(x,y,t)$, expressed as:

$$\mathscr{G}_\theta : h(x,y) \mapsto \widetilde{w}(x,y,t), \tag{12}$$

where the parameters θ represent the learnable weights of the model. Through spectral convolution in the Fourier domain, the network captures long range wave propagation characteristics and dispersive behavior, enabling an end-to-end prediction of the dynamic wavefield governed by Kirchhoff-Love equation. The overall framework is shown in Fig. 1(c), comprising convolutional layer 1, FNO layer, convolutional layer 2, UFNO layer, and convolutional layer 3.

The first convolutional layer (Conv layer 1) uses two identical 3×3 convolutions and a Rectified Linear Unit (ReLU) activation to transform a single channel input thickness map into a multi-channel feature representation. This step expands the feature dimension and extracts essential spatial patterns from the geometric input, providing a rich representation for the subsequent spectral learning stage.

The FNO layer containing two identical FNO blocks operates in frequency domain to model long range interactions. Key hyperparameters include a channel width of 64 for FNO blocks and a Fourier mode truncation of (32, 32) in frequency domain. Each block applies a 2D FFT to the feature tensor, truncates the high frequency modes, and retains only the first (32, 32) low frequency Fourier modes to reduce computational costs. The retained spectral coefficients are multiplied by a learnable complex valued matrix R, which acts as a trainable filter in frequency domain. The transformed features are then mapped back to the spatial domain through an inverse FFT and combined with a 1×1 convolution. As shown in Fig. 1(a), The output of one FNO block is therefore given by:

$$l_{k+1}(x) = \sigma\left(W \cdot l_k(x) + \mathscr{F}^{-1}(R \cdot \mathscr{F}[l_k(x)])\right), \tag{13}$$

where $\mathscr{F}$ and $\mathscr{F}^{-1}$ denote the Fourier and inverse Fourier transforms, W is a linear transformation applied in the spatial domain, and $\sigma(\cdot)$ is the ReLU activation function. Two such FNO blocks are sequentially connected to strengthen the spectral modeling capability and enhance numerical stability.

Following the FNO layer, a convolutional layer (Conv layer 2) refines the fused feature maps and mitigates potential spectral artifacts by performing local spatial filtering.

The next UFNO layer consists of a symmetric encoder-decoder structure with four levels of downsampling and upsampling, forming a multi-resolution hierarchy that combines the global frequency learning of FNO with the multi-scale feature extraction of UNet. As shown in Fig. 1(b), in each encoding stage, an FNO block first extracts global information at the current scale, followed by a 3×3 convolution for downsampling, accompanied by a ReLU nonlinearity. The output of each encoder stage is simultaneously passed through a 1×1 convolution to generate a skip-connection tensor that preserves fine scale details for later fusion. Consequently, the spatial resolution of the feature maps gradually decreases, allowing the network to accumulate both global and local features in a compact latent space. In the decoding path, the process is reversed through four corresponding stages of upsampling. Each decoder stage starts with a transpose convolution to double the spatial resolution, followed by a ReLU activation and another FNO block that reconstructs the global dependencies lost during downsampling. At each resolution, the upsampled features are added to the corresponding skip-connection tensor from the encoder to fuse multi-scale information. Because the combination of convolution and transpose convolution operations may lead to one pixel misalignment between the encoder and decoder feature maps, a bilinear interpolation is applied to ensure perfect spatial alignment before the fusion. This design allows UFNO to account for long range wave propagation while preserving localized spatial variations, achieving accurate reconstruction of scattering wavefield.

A final 1×1 convolutional layer (Conv layer 3) converts the feature tensor into the spatiotemporal output wavefield after UFNO. Each channel of the resulting prediction $\widetilde{w}(x,y,t)$ represents the out-of-plane displacement distribution at a specific moment.

During training, the network parameters θ are optimized to minimize the mean squared error (MSE) between the predicted $\widetilde{w}_{i,j}^n$ and reference wavefields $w_{i,j}^n$ obtained from finite difference calculations:

$$\mathscr{L}_{data} = \frac{1}{N_{data}} \sum_{i,j,n} \left(\widetilde{w}_{i,j}^n - w_{i,j}^n\right)^2, \tag{14}$$

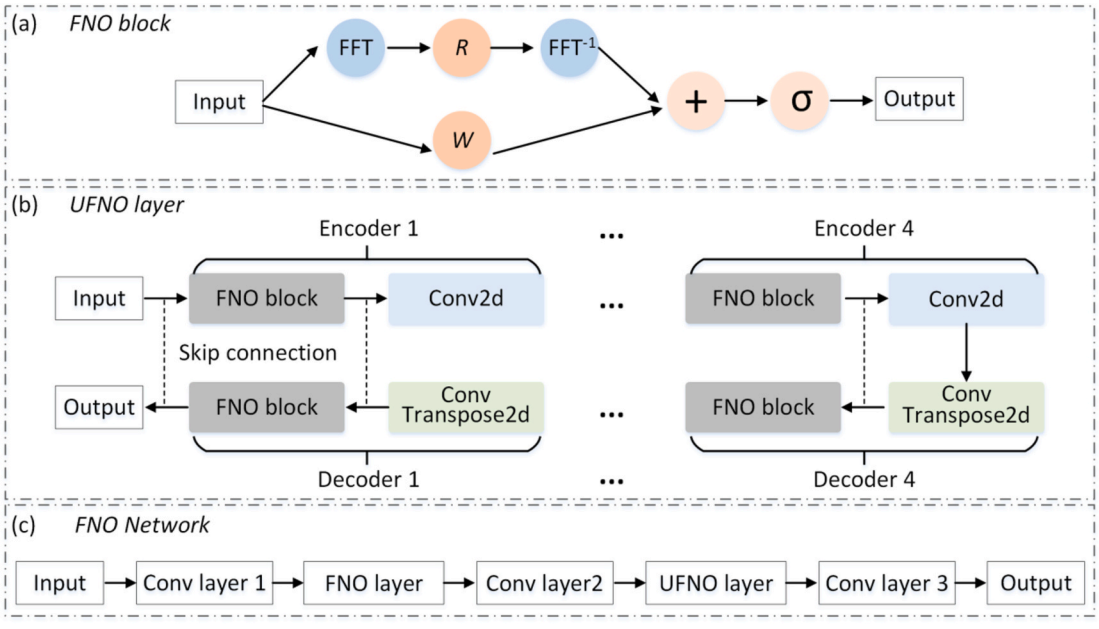


Fig. 1. FNO network architecture.

where N_{data} denotes the total number of interior grid points considered in both space and time. The optimization is performed using the Adam algorithm. The FNO network serves as the baseline framework, upon which physics embedded and residual refinement are subsequently incorporated.

2.4. Residual evaluation of Kirchhoff-Love equation

Although the FNO network approximates the mapping from plate thickness to the corresponding wavefield, the prediction does not inherently guarantee strict compliance with Kirchhoff-Love equation. To quantify the physical consistency of the predicted wavefield, a residual evaluation procedure is performed by substituting the predicted wavefield into the discrete Kirchhoff-Love equation. This process enables direct measurement of the discrepancy between the predicted dynamics and the theoretical equation of motion.

Given a spatiotemporal displacement field $\widetilde{w}_{i,j}^{n}$ at time n and coordinates (i,j) obtained from the FNO network, its physical consistency with Kirchhoff-Love equation can be assessed by substituting it back into the same finite difference formula used for forward modeling. From Eq. (6), the discrete Kirchhoff-Love equation at node (i,j,n) reads:

$$\frac{\widetilde{w}_{i,j}^{n+1}-2\widetilde{w}_{i,j}^{n}+\widetilde{w}_{i,j}^{n-1}}{(\Delta t)^2}-\frac{f\left[\widetilde{w}_{i,j}^{n}\right]}{\rho h(i,j)}=\frac{q_{i,j}^{n}}{\rho h(i,j)}, \tag{15}$$

where $f\left[\widetilde{w}_{i,j}^{n}\right]$ is obtained through Eq. (10). When the external excitation $q_{i,j}^{n}$ is negligible in the observation region, the point-wise residual of Kirchhoff-Love equation is therefore defined as:

$$r_{i,j}^{n}=\frac{\widetilde{w}_{i,j}^{n+1}-2\widetilde{w}_{i,j}^{n}+\widetilde{w}_{i,j}^{n-1}}{(\Delta t)^2}-\frac{f\left[\widetilde{w}_{i,j}^{n}\right]}{\rho h(i,j)}. \tag{16}$$

The magnitude of $r_{i,j}^{n}$ represents the instantaneous local violation of Kirchhoff-Love dynamics, reflecting how closely the prediction aligns with the true physical behavior. If the wavefield is an exact numerical solution generated by the same finite difference operator, $r_{i,j}^{n}$ approaches zero. The global residual energy is defined as the MSE of all local residuals within a selected solution domain:

$$\mathscr{L}_{PDE}=\frac{1}{N_{PDE}}\sum_{i,j,n}\left[\frac{\widetilde{w}_{i,j}^{n+1}-2\widetilde{w}_{i,j}^{n}+\widetilde{w}_{i,j}^{n-1}}{(\Delta t)^2}-\frac{f\left[\widetilde{w}_{i,j}^{n}\right]}{\rho h(i,j)}\right]^2, \tag{17}$$

where N_{PDE} denotes the total number of interior grid points considered in both space and time, excluding the boundary nodes to eliminate edge effects. This residual analysis provides a unified and quantitative way to assess the physical fidelity of the predicted wavefields.

3. Numerical simulations and results

3.1. Dataset construction

The dataset used in this study is generated through modeling of Lamb wave propagation in plate structures containing random defects. Fig. 2 shows the wave propagation model and the group velocity dispersion curve of this model. The computational domain corresponds to an aluminum plate with dimensions of 500 mm × 500 mm × 3 mm. Defects are introduced in the form of locally reduced plate thickness, and their spatial ranges are restricted to a circular region centered at (250 mm, 250 mm) with a radius of 60 mm. For each defect sample, both the geometric configuration and the thickness distribution are generated randomly. When characterized using an equivalent circular metric, the defect radius ranges from 10 mm to 40 mm, with an unfixed center position. The remaining plate thickness within the defect region varies between 1.5 mm and 3 mm, whereas the undamaged area retained the nominal thickness of 3 mm. These settings ensured sufficient variability and stochasticity in the defect morphology, thereby enhancing the dataset's representativeness for practical scenarios. A total of 1280 defect samples are generated. Each thickness map is discretized on a 501 × 501 grid with a spatial resolution of 1 mm. To reduce data dimensionality, downsampling with a stride of two was applied along both spatial directions. The resulting dataset, consisting of 1280 thickness maps of size 251 × 251, serves as the input to the FNO framework. Fig. 3 shows the thickness maps of several defects in the dataset.

For each defect sample, the corresponding wavefield was obtained by solving Kirchhoff-Love equation using a finite difference scheme. Specifically, the solution domain represents a square aluminum plate of 500 mm × 500 mm, discretized with spatial intervals of $\Delta x = \Delta y = 1\ mm$, resulting in $N_x = N_y = 501$ grid points in each direction. The temporal increment is set to $\Delta t = 0.01\ \mu s$, and the total simulation duration is 150 μs, corresponds to $N_t = 15001$ time steps. These settings ensure sufficient spatial and temporal resolution for capturing Lamb wave propagation within the stability constraint of the explicit scheme. The excitation source is a five-cycle sinusoidal burst signal with a center frequency of 200 kHz, modulated by a Hanning window, located at (250 mm, 50 mm). For the finite difference solver adopted in this work, it takes 7.27 h to generate the complete dataset on a workstation equipped with an NVIDIA RTX A5000 GPU. For each thickness distribution, after obtaining the complete wavefield data through finite difference modeling, the data were spatially sampled at 2 mm intervals and temporally sampled every 0.5 μs within the time window of 25 μs to 125

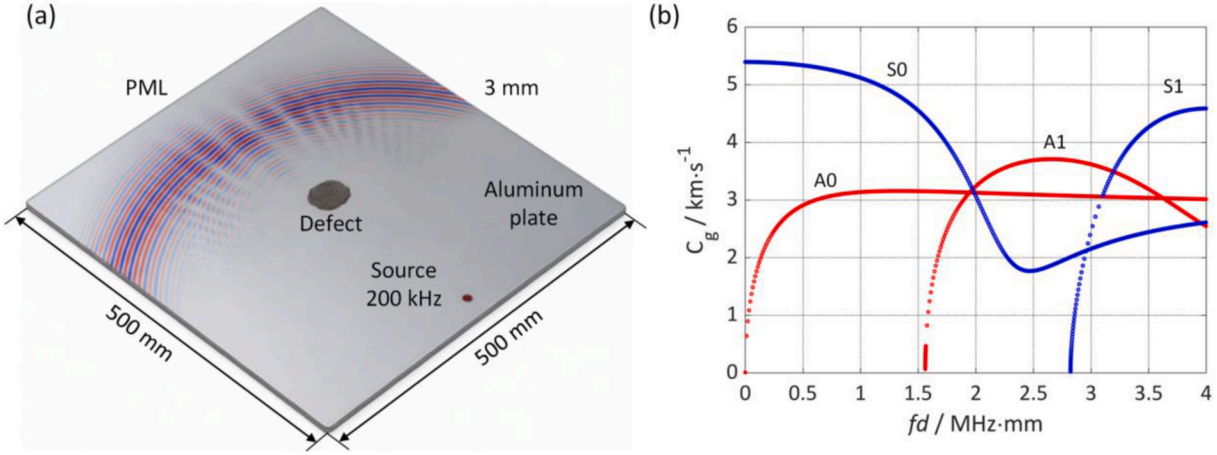


Fig. 2. (a). Wave propagation model. (b) Group velocity dispersion curve.

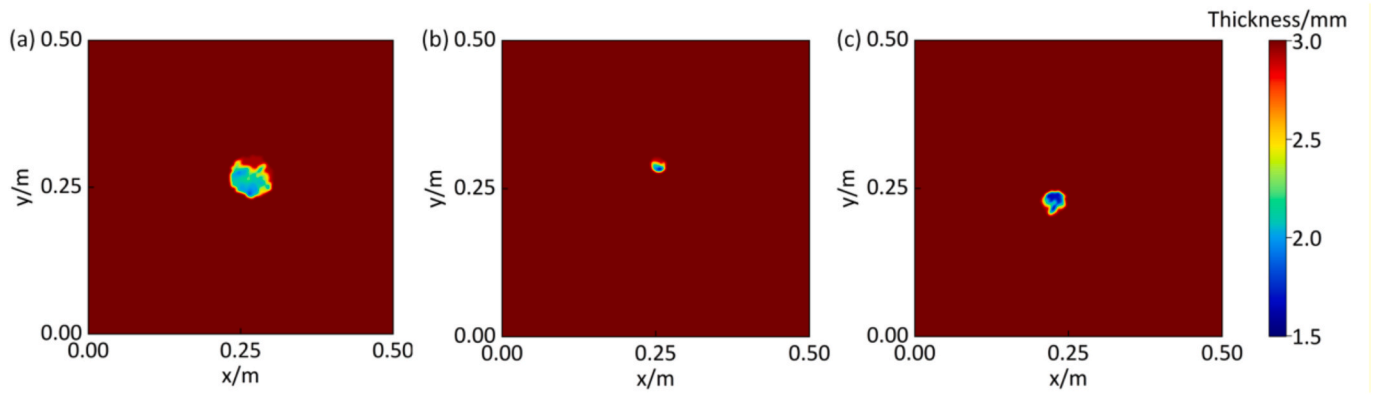


Fig. 3. Defect samples in dataset.

μs. The resulting wavefield dataset thus has a dimension of 1280 × 251 × 251 × 201 in (x,y,t), serves as the output to the FNO framework.

To validate the reliability of the dataset used in this study, 3D wavefields for both defect-free and an irregular defect were computed using a commercial finite element software, from which the out-of-plane displacement fields were extracted. These results were compared with the wavefields obtained using the finite difference scheme based on Kirchhoff-Love plate equation adopted in this work. The agreement between the two sets of results confirms the validity and physical fidelity of the constructed dataset. Detailed comparison results are provided in Appendix.

3.2. Training and wavefield prediction

We trained and evaluated both the baseline FNO framework shown in Fig. 1 and the proposed enhanced variant dual-stage FNO with physics embedded and residual refinement. The wavefield predictions were subsequently compared and analyzed. Specifically, for baseline FNO, thickness maps serve as input and the corresponding wavefields as output, with the network optimization governed by Eq. (14). This baseline configuration is referred to as Stage 0. Keeping all other settings in Stage 0 unchanged, Kirchhoff-Love equation was incorporated into the loss function through Eq. (17) as the PDE loss term. This term was combined with the data loss through an explicit weighting scheme to form the total training loss, which jointly constrained the network optimization. This procedure is referred to Stage 1. The total training loss function is expressed as:

$$\mathscr{L}_{total} = \mathscr{L}_{data} + \lambda \mathscr{L}_{PDE}, \tag{18}$$

where λ serves as the weight parameter to adjust the constraint effect of data loss and PDE loss, optimizing the iteration of FNO. The weighting parameters is critical to the effectiveness of physics embedding. Improper values may fail to improve the learning performance and can even degrade it. As the predictive accuracy increases, the discrepancy between the predictions and references decreases, we examined Stage 1 data loss under different λ, as shown in Fig. 4, and selected $\lambda = 0.1$ as the most appropriate choice.

The learning rate for both Stage 0 and Stage 1 was set to 1×10^{-3}, and each framework was trained for 1000 iterations. After Stage 1, the same network architecture is used to capture the remaining discrepancy. The input is formed by concatenating the thickness maps with the Stage 1 predictions, and the output is the residual wavefields between Stage 1 predictions and the references. The final wavefield predictions are expressed as the superposition of this outputs and the Stage 1 outputs. This process is defined as Stage 2. Since Stage 2 focuses on residual refinement, several network parameters were adjusted. Specifically, the number of retained Fourier modes in FNO blocks was increased from (32,32) to (64,64) to capture high frequency information, the learning

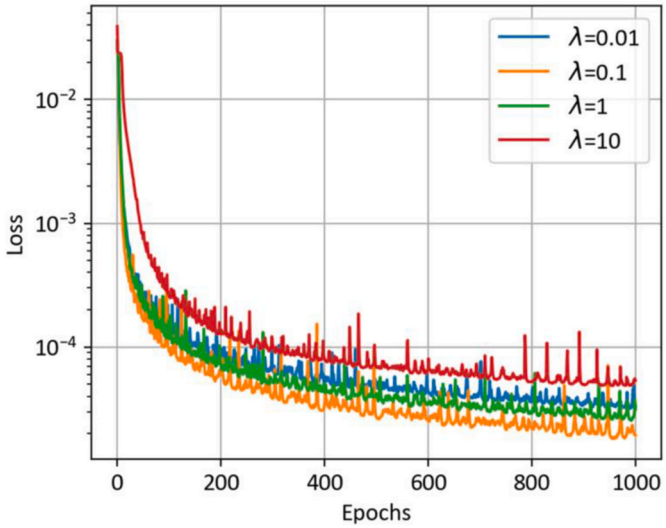


Fig. 4. Influence of different weight parameters on convergence.

rate was set to 1×10^{-4}, and the number of training iterations was reduced to 500. An early stopping strategy was also employed to prevent overfitting. On the same workstation, for the training set of 1024 samples, the training times for Stage1 and Stage2 were 23.59 h and 13.52 h respectively. Therefore, the total training time can be regarded as the sum of the two, which is 37.11 h. For the complete dataset of 1280 samples, the total prediction time was 287 s, among which the prediction times for Stage1 and Stage2 were 206 s and 81 s respectively. It should be noted that, prior to Stage 2 training, maximum normalization was applied to scale the residual wavefields to approximately [−1, 1]. This preprocessing step enhances training stability, as residual wavefield amplitudes are smaller than those of the original wavefields following Stage 1. During inference, the Stage 2 output (normalized residual) is denormalized using the identical scaling factor derived from the training set, thereby restoring the residual wavefield to its original physical units. The final predicted wavefield is then computed as the sum of this denormalized residual and the Stage 1 prediction. Crucially, this dual-stage reconstruction preserves physical consistency and ensures end-to-end fidelity in the predicted wavefield's amplitude and scale.

Fig. 5 shows the training loss histories. Fig. 5(a) compares the data losses of Stage 0 and Stage 1 on the training set, together with the evolution of PDE loss introduced in Stage 1, while Fig. 5(b) presents the training and testing data losses in Stage 2. As illustrated in Fig. 5(a), the baseline Stage 0 model exhibits a rapid initial decrease in data loss, followed by a slower convergence accompanied by slight fluctuations throughout the training process. After incorporating Kirchhoff-Love

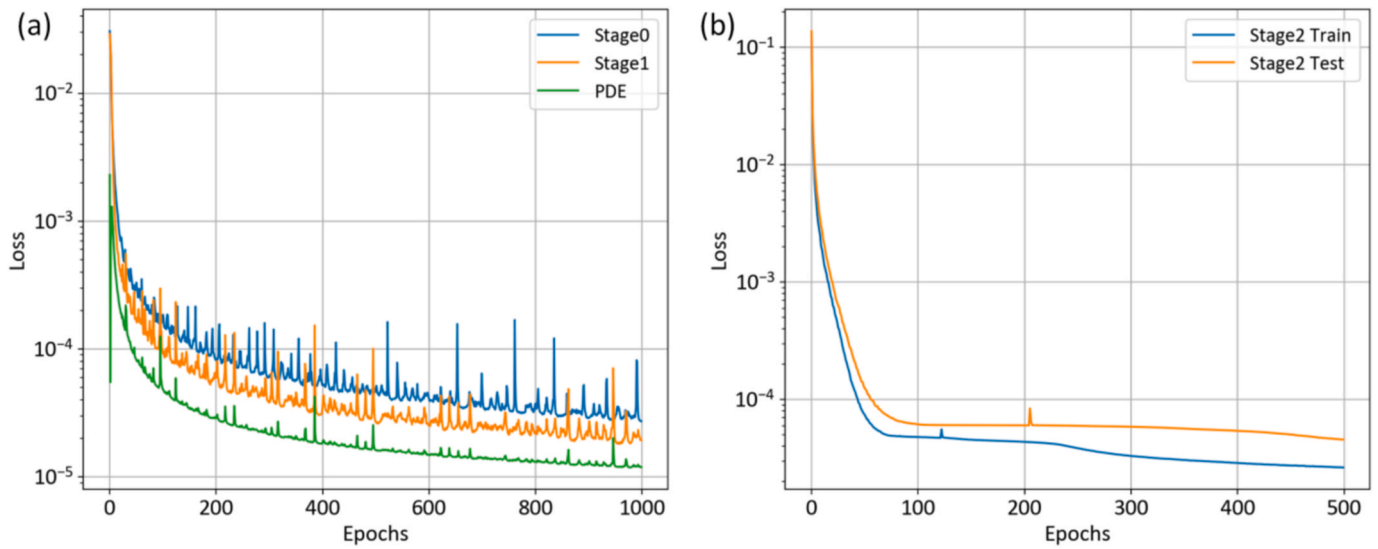


Fig. 5. Training loss curves.

equation as an additional physical constraint in Stage 1, the data loss remains consistently lower than that of Stage 0 during the entire convergence process. This behavior indicates that the physics constraint effectively regularizes the optimization procedure and suppresses non-physical fluctuations during training. Meanwhile, the PDE loss decreases steadily and eventually reaches a stable level, suggesting that the predicted wavefields progressively satisfy the governing equation while remaining consistent with the training data. Fig. 5(b) illustrates the residual refinement process in Stage 2. Both training and testing losses decrease rapidly in early epochs and then converge smoothly. The discrepancy between the training and testing loss curves remains limited, confirming that the residual learning strategy does not lead to severe overfitting. The physics embedding introduced in Stage 1 guides the network toward physically consistent solutions, while Stage 2 further reduces the prediction error and refines local discrepancies that may be difficult to capture within Stage 1 framework, without sacrificing generalization capability.

After training, we validated and compared the wavefield predictions of Stage 0 and Stage 2 for the defect shown in Fig. 3(a). Fig. 6 compares the prediction wavefield snapshots at several representative time instants t = 25, 50, 75, 100, and 125 μs. The first row shows the reference wavefields obtained from finite difference modeling, while the second and third rows present the Stage 0 predictions and their corresponding errors relative to the references. The fourth and fifth rows display the Stage 2 predictions and the associated error distributions. At early times (t = 25 and 50 μs), when the wave has not yet interacted with the defect, both Stage 0 and Stage 2 capture the overall propagation pattern reasonably well. The predicted wavefronts exhibit good agreement with the references, indicating that the wave propagation behavior can be learned by a purely data-driven framework. Small but discernible discrepancies are still visible in the error distributions, particularly near the source and along the circular wavefronts. However, Stage 2 has shown a smaller error distribution than Stage 0. As wave propagates further and interacts with the defect region (t = 75 and 100 μs), the differences between the two frameworks become more pronounced. In Stage 0, noticeable error concentrations emerge around the defect induced scattering zone, and the error magnitude increases as the wavefield becomes more complex. These errors reflect the difficulty of a purely data-driven model in accurately representing scattering phenomena. In contrast, the Stage 2 predictions remain closely aligned with the reference wavefields, and the corresponding error maps show a clear reduction in both amplitude and spatial range, especially near the defect. At a later time (t = 125 μs), when multiple scattered wave components exist and the wavefield exhibits increased complexity, the advantage of the proposed strategy becomes even more evident. While Stage 0 suffers from accumulated errors distributed over a wide region, Stage 2 maintains a substantially lower error level and preserves more scattering features. Fig. 6 intuitively demonstrates the improvement of Lamb wave scattering prediction using the proposed dual-stage FNO strategy based on physics embedded and residual refinement.

Fig. 7 further compares the predictions and references from the perspective of local time-domain signals and wavefield directivity, shifting the focus from global spatial wavefields to localized temporal responses and propagation characteristics. After obtaining the predicted wavefields, monitoring points were uniformly sampled along a circular trajectory with a radius of 150 mm, centered at (250 mm, 250 mm). A total of 360 points were collected at 1° intervals, yielding 360 signals corresponding to all azimuthal directions.

Fig. 7(a) shows representative signals extracted every 30°, resulting in 12 angularly distributed signals for comparison. The stacked signals exhibit an S-shaped pattern, which is attributed to the angular variation of propagation distance from the excitation point located at (250 mm, 50 mm). As the azimuthal angle changes, the travel distance to the monitoring points along the circular path varies accordingly, leading to systematic shifts in signal arrival times. Across all angles, the Stage 2 predictions closely match the reference signals in terms of arrival time and waveform shape. The phase alignment between the predicted and reference signals is well preserved, and the signal amplitudes show only minor deviations, even in directions where scattering effects are pronounced. This agreement indicates that the proposed framework accurately captures both the temporal evolution and angular dependence of the scattering waves at local observation points. To further quantify the directional characteristics, the maximum amplitude of each signal was extracted and mapped onto a polar coordinate system after normalization, as shown in Fig. 7(b). The resulting energy distribution pattern predicted by Stage 2 exhibits good agreement with the reference over the full 360° range. Both the dominant radiation direction and the overall angular distribution are well reproduced, including local peaks related to scattering. The close overlap between the two curves demonstrates that the proposed dual-stage FNO approach not only reconstructs the local waveform accurately but also preserves the anisotropic energy redistribution caused by the defect.

3.3. Error analysis and statistical evaluation

To quantitatively evaluate the effectiveness of the proposed dual-

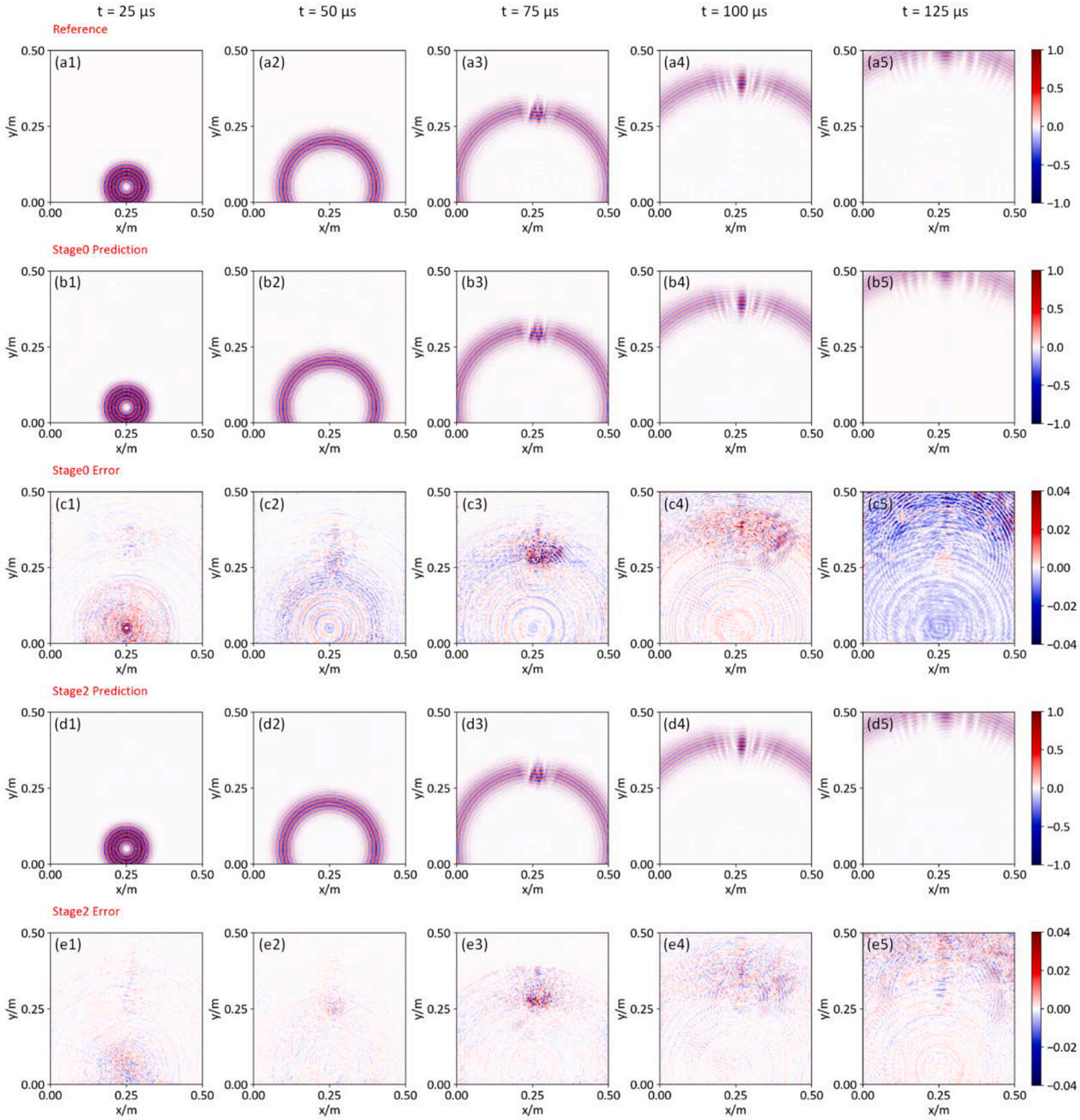


Fig. 6. Comparison of wavefield predictions of the defect shown in Fig. 3(a).

stage FNO strategy, a statistical error analysis was conducted on both training and testing datasets. For each spatial location in computational domain, a time-domain signal of length 201 was obtained. The relative ℓ_2 error between the predicted signal and the reference solution was calculated for Stage 0, Stage 1, and Stage 2, and used as the primary error metric. For a specific sample, the average error of the signals at all points is taken as the error of that sample. The relative ℓ_2 error is defined as

$$\varepsilon = \frac{\left\| s_{pred} - s_{ref} \right\|_2}{\left\| s_{ref} \right\|_2}, \tag{19}$$

where s_{pred} and s_{ref} denote the predicted and reference signals at a given spatial location, respectively.

Fig. 8 presents the comparison of relative ℓ_2 error over all spatial locations, separately for the training and testing datasets. The baseline Stage 0 model exhibits relatively large median errors and broad error dispersion, indicating substantial spatial variability. With the introduction of physics embedded learning in Stage 1, both the median error and error spread are reduced on the training and testing datasets. Further error reduction is achieved in Stage 2 through residual refinement, which yields the lowest median errors and the most compact distributions across both datasets, demonstrating that the dual-stage FNO strategy enhances accuracy while maintaining generalization

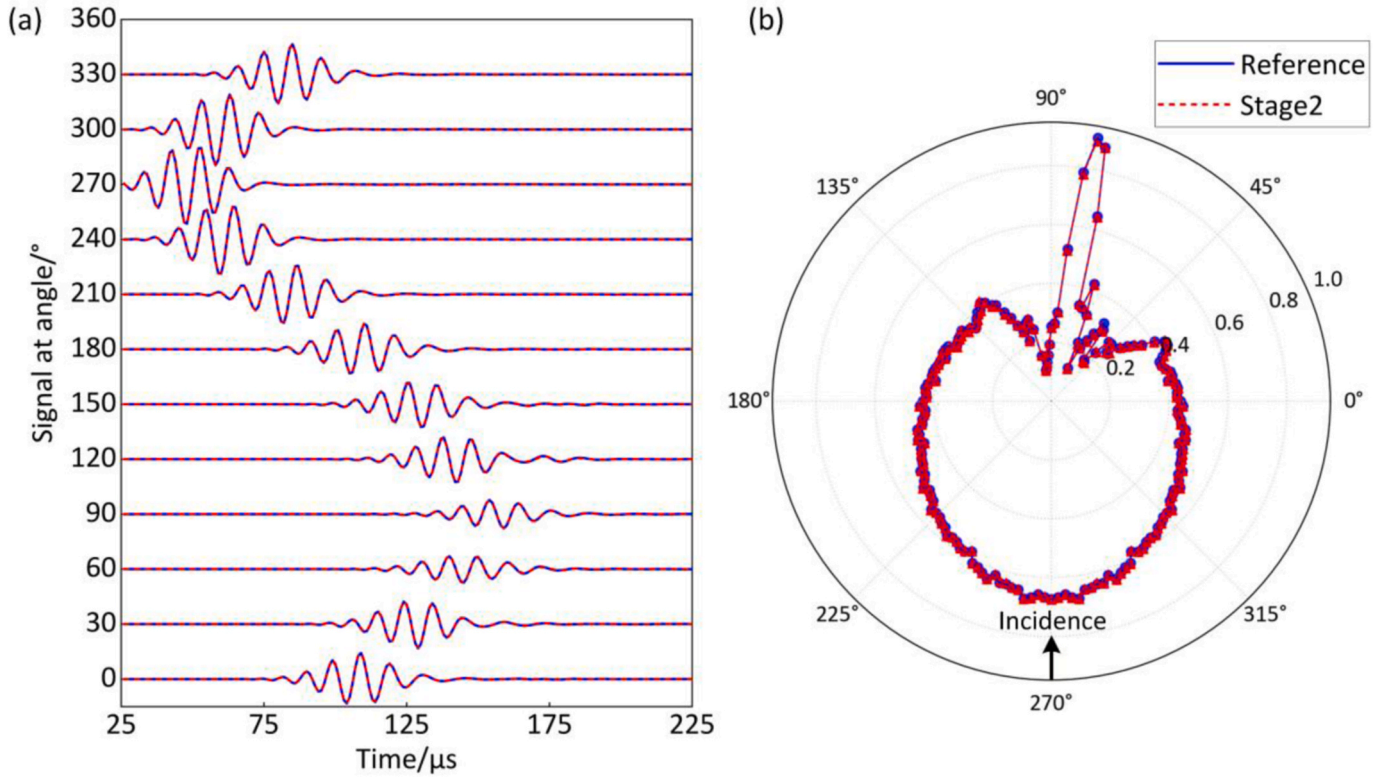


Fig. 7. Comparison of signal predictions and the energy distribution pattern (the black arrow indicates the direction of incidence).

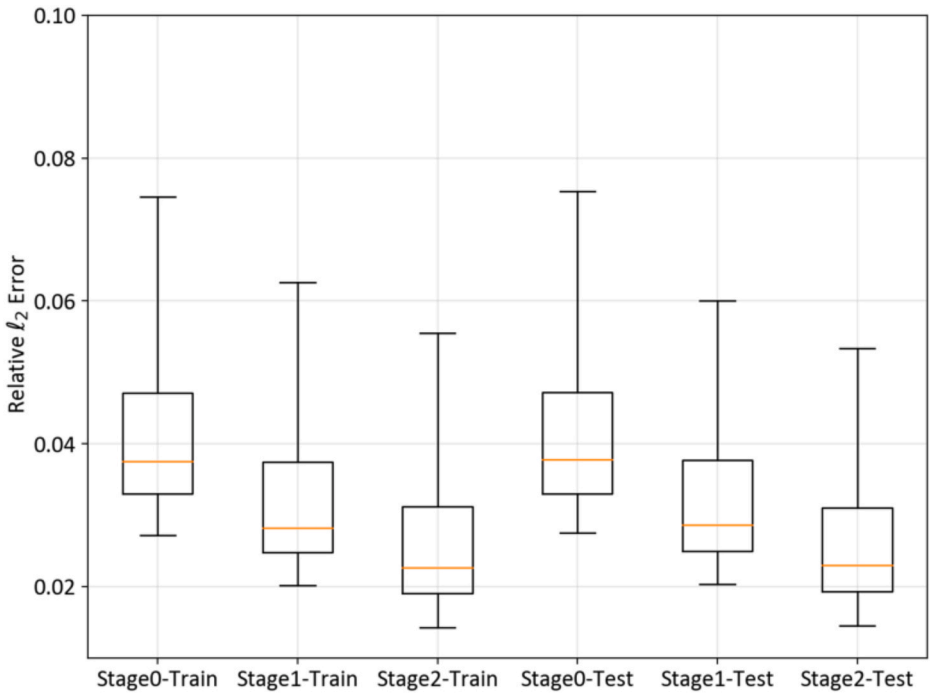


Fig. 8. Distribution of relative ℓ_2 errors of Stage 0, Stage 1, and Stage 2.

capability.

Fig. 9 provides a more direct assessment of performance gains by showing the relative improvement of Stage 1 and Stage 2 with respect to the baseline Stage 0. The improvement is defined as the percentage reduction in relative ℓ_2 error compared to Stage 0, and the distributions are reported separately for the training and testing datasets. As shown in Fig. 9, Stage 1 provides a consistent improvement over the baseline, with most samples exhibiting error reductions of approximately 15% to 30%, indicating a systematic enhancement across the dataset. Stage 2 achieves substantially larger gains, with improvements typically ranging from 30% to 45% for both training and testing samples. Together, physics embedded and residual refinement act in a complementary

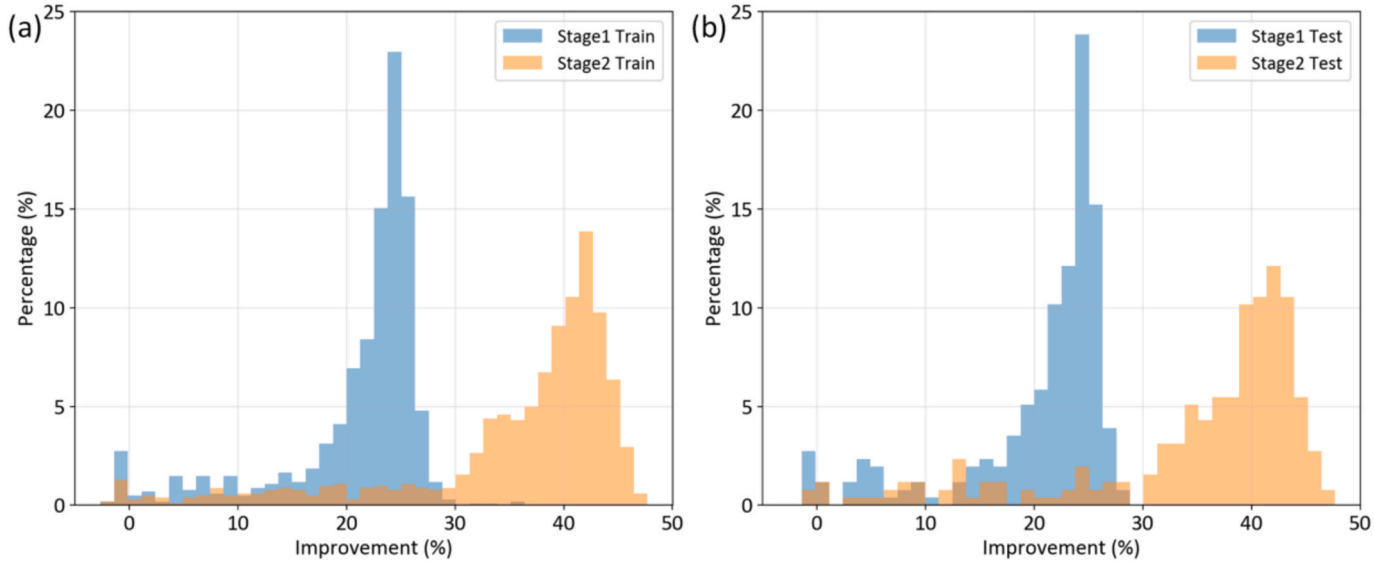


Fig. 9. The improvement of Stage1 and Stage2.

manner, with the former reducing global errors and enhancing physical consistency, and the latter further correcting localized discrepancies, thereby yielding a more accurate and robust predictive model.

Fig. 10 presents the temporal evolution of the relative ℓ_2 error of the predicted wavefield snapshots, where the error at each time step is defined as the average over all spatial samples and the shaded bands denote one standard deviation for the training and testing datasets. For all three frameworks, the relative ℓ_2 error increases with time as wave propagation and scattering lead to progressively more complex wavefield patterns, yet clear performance differences are observed. The baseline Stage 0 model exhibits the largest error throughout the entire time interval. Incorporating physical constraints in Stage 1 leads to a consistent reduction in error across all time steps and improve the physical consistency of the predicted wavefield during propagation. The lowest error levels are achieved by Stage 2 through residual refinement. The similar trends observed in the training and testing datasets indicate the performance improvements are not a result of overfitting, but rather reflect a true enhancement in generalization capability. These results further confirm that the proposed dual-stage FNO consistently provides more accurate and reliable wavefield predictions for Lamb wave scattering caused by complex defects.

4. Experiments and results

4.1. Experiment setup

Scanning experiments were conducted using the laser Doppler vibrometer Polytec PSV-500 to validate the effectiveness and practical applicability of the proposed method for predicting Lamb wave scattering wavefields. To minimize the influence of boundary reflections, a 6061 aluminum alloy with dimensions of 1000 mm $\times$ 1000 mm $\times$ 3 mm was employed, which is larger than the size of the computational domain in finite difference modeling. As is shown in Fig. 11, an irregular defect with randomly distributed thickness was machined at the center of the plate, located at (500 mm, 500 mm). The minimum remaining thickness within the defect region was measured to be 1.98 mm. A piezoelectric transducer (PZT) with a center frequency of 200 kHz was bonded at (500 mm, 300 mm), corresponding to a distance of 200 mm

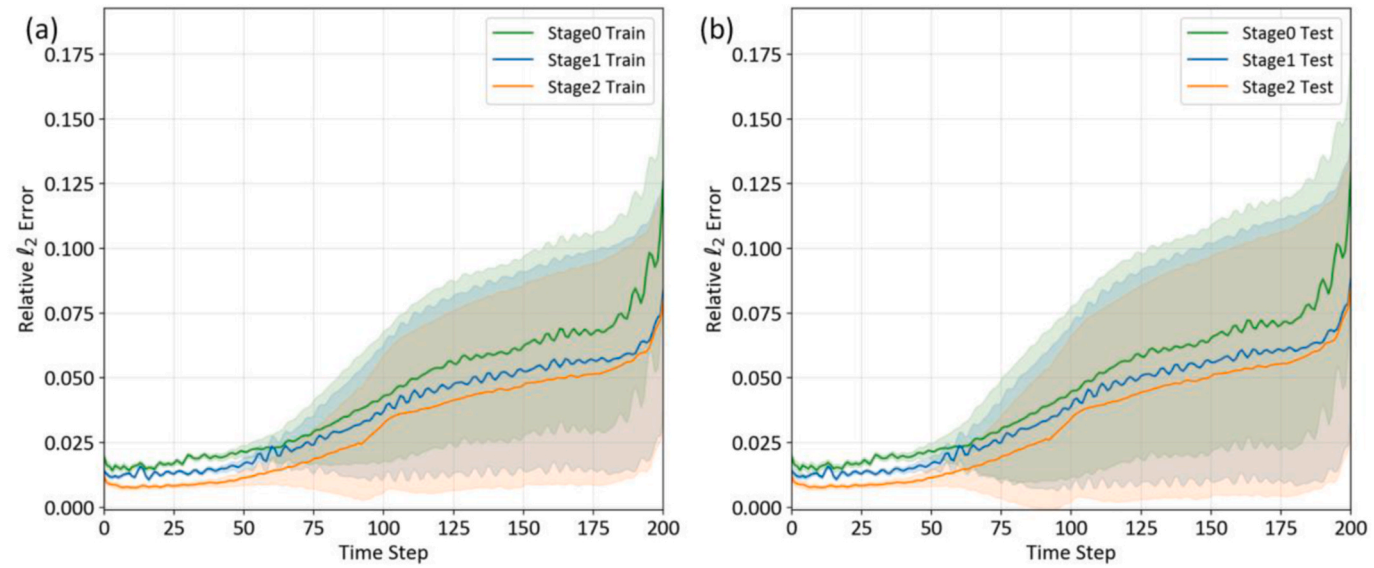


Fig. 10. Distribution of relative ℓ_2 errors of predicted wavefield at different moments.

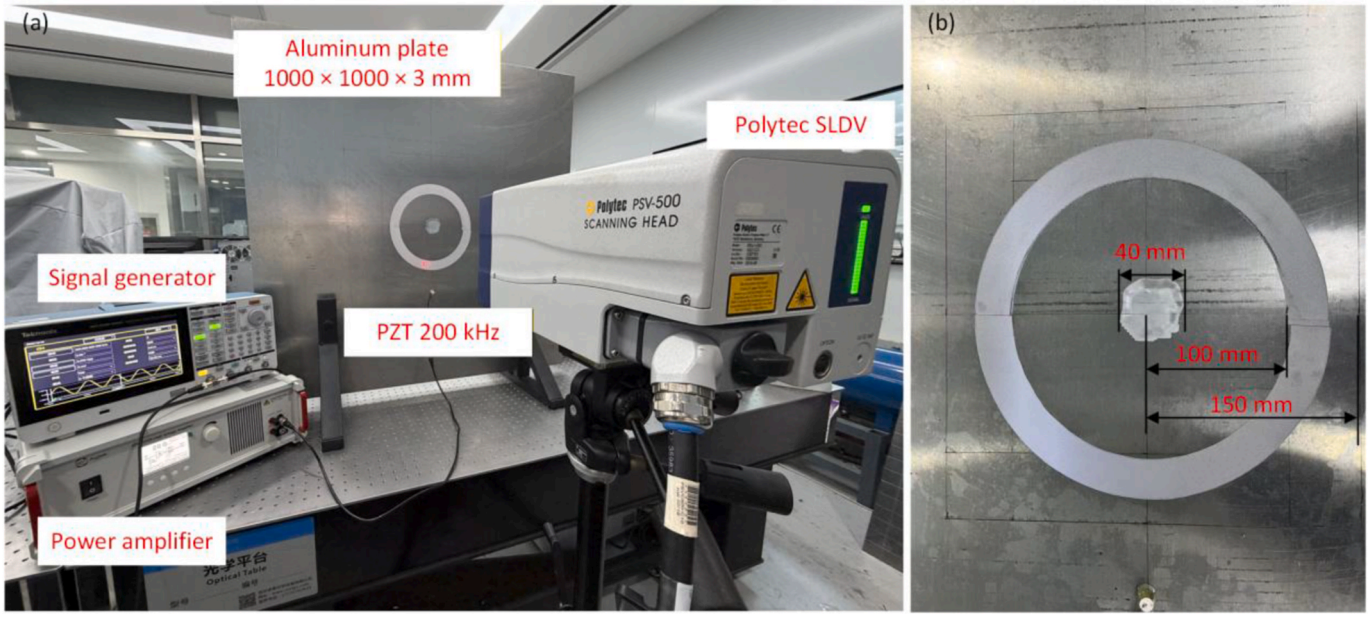


Fig. 11. Laser scanning vibration measurement experiment.

from the plate center, consistent with the excitation configuration adopted in numerical simulations. A five-cycle sinusoidal burst signal with a center frequency of 200 kHz and a peak-to-peak voltage of 3 V was generated using a Tektronix AFG 31000 arbitrary function generator. The excitation signal was subsequently amplified by an Aigtek ATA-2022B dual-channel power amplifier with a gain of 20 dB and then applied to the PZT. During the measurements, the Polytec scanning head and the plate were carefully aligned to ensure that the laser beam was as close to normal incidence to the plate surface as possible. The scanning region was defined as a circular trajectory centered at the defect, with a radius of 110 mm. A total of 360 measurement points were uniformly distributed along the trajectory at 1° intervals. To enhance the signal to noise ratio and improve the vibration detection capability, reflective materials were attached along the scanning path. The scanning sampling frequency was set to 6.25 MHz with a record length of 2500 points. A band-pass filter ranging from 150 kHz to 250 kHz was applied to isolate the frequency components of interest. At each measurement point, 50 repeated acquisitions were performed, and the averaged signal was used for subsequent analysis.

4.2. *Wavefield and signal analysis*

We first compared the wavefield predictions of Stage 0 and Stage 2 for the defect fabricated in the experiment. As shown in Fig. 12, the prediction results of Stage2 exhibit smaller errors in both amplitude and spatial distribution. As wave propagates further, passes through the defect, and undergoes scattering, the advantage of the proposed dual-stage framework becomes more evident. This demonstrates the improvement in Lamb wave scattering prediction achieved by the proposed dual-stage FNO strategy with physics embedding and residual refinement.

Fig. 13(a) presents the time-domain signals scanned along the circular trajectory, with representative waveforms displayed at every 30° intervals from 360 angular sampling positions. Similar to the numerical results, the experimental signals exhibit a clear S-shaped temporal distribution, which arises from the variation in propagation distance between the excitation source and the scanning points along the circular trajectory. Compared with numerical simulations, the experimental waveforms display more pronounced dispersive characteristics. The wave packets become progressively broadened as the propagation distance increases. This behavior reflects the inherent dispersive characteristics of A0 Lamb mode within the selected frequency range, as well as additional effects introduced by real experimental conditions, including material damping, surface roughness, and slight excitation bandwidth broadening. Fig. 13(b) compares the energy distribution pattern obtained from the reference wavefield, experimental measurements, and Stage 2 predictions. The directivity result is obtained from the normalized maximum amplitude of the signal at each angular position. The Stage 2 results show good agreement with both the reference solution and experimental measurements in terms of the main radiation direction and overall angular distribution. In particular, the dominant lobes and the asymmetric energy redistribution induced by the defect are well captured. Minor amplitude discrepancies appear at certain angles, which may be attributed to experimental noise and unmodeled factors such as material attenuation and measurement uncertainty. Nevertheless, the close correspondence between measured and predicted directivity patterns demonstrates the effectiveness of the proposed dual-stage FNO under realistic experimental conditions.

Fig. 14 shows the continuous wavelet transform (CWT) analysis of the scanning signal at 90° direction. This signal corresponds to the longest propagation path along the circular trajectory and therefore exhibits the strongest dispersive behavior. The time-frequency results show that the signal energy is concentrated around 200 kHz and has a certain bandwidth. This frequency spreading is partly attributed to the dispersive nature of A0 mode in the considered frequency-thickness range. As wave propagates and interacts with the defect, different frequency components travel at different group velocities, resulting in temporal separation and frequency dependent energy redistribution. Consequently, the wave packet undergoes a certain degree of distortion, with multiple frequency components coexisting at a given moment. This behavior is more evident in experiments due to additional factors such as material attenuation and measurement noise. Moreover, the consistency between the time-frequency energy distribution and the theoretical A0 dispersion curve confirms the dominance of A0 mode in the measured signal and the correct representation of its dispersive behavior. This consistency further supports the physical fidelity of both the experimental measurements and the proposed dual-stage FNO with physics embedded and residual refinement.

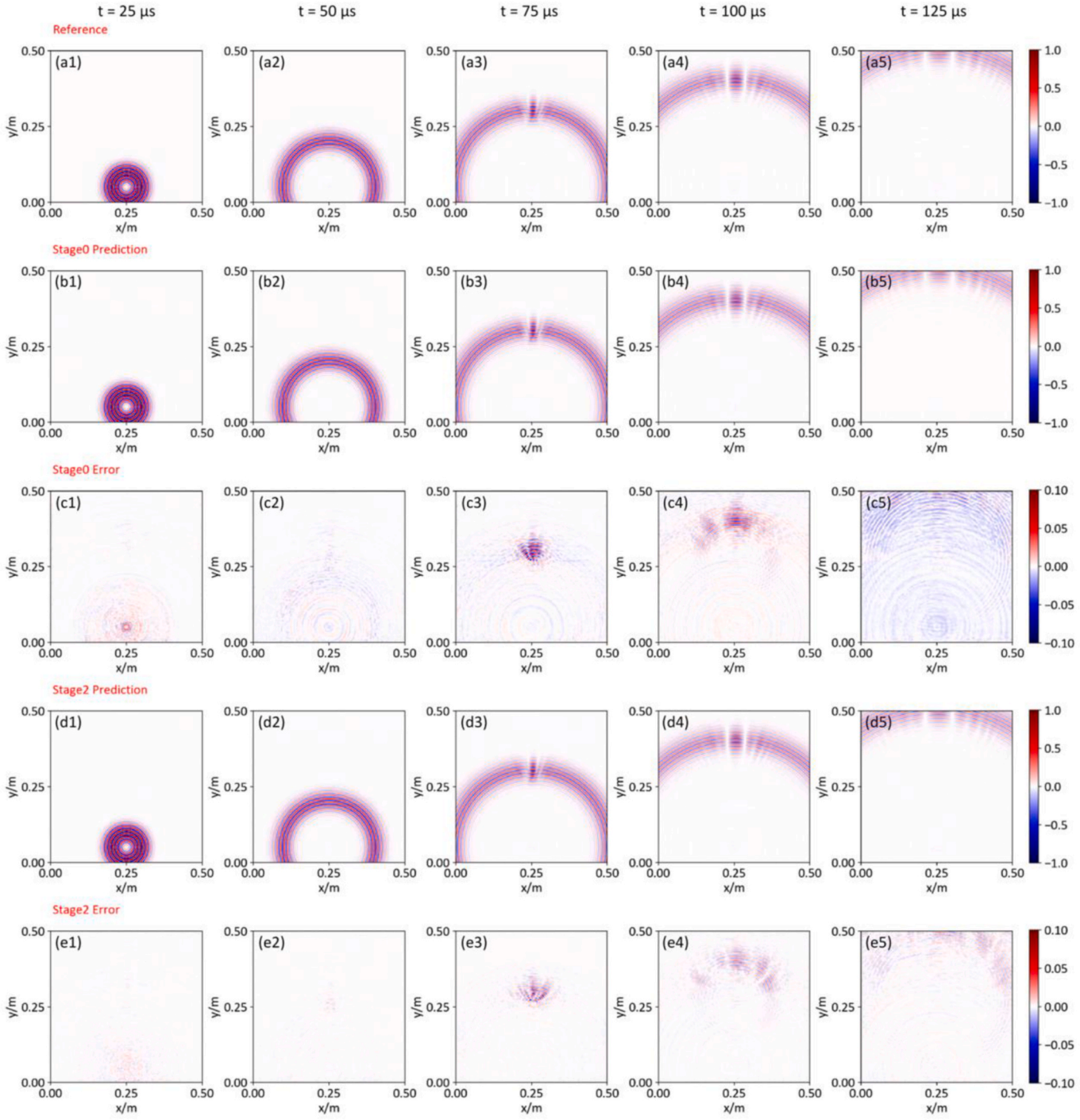


Fig. 12. Comparison of wavefield predictions of the defect in experiment.

5. Discussion

5.1. Key features and advantages

The proposed dual-stage Lamb wave scattering modeling framework with physics embedded and residual refinement integrates the complementary strengths of PINN and FNO. By learning a direct operator mapping from structural parameters to full spatiotemporal scattering wavefields, the framework avoids the configuration specific nature and pointwise coordinate dependence commonly associated with PINN based approaches. It demonstrates the ability to generalize over a family of related governing problems, as reflected in its stable performance across scattering problems involving randomly varying defect locations, geometries, and thickness distributions. Meanwhile, embedding governing equations into the training process guides the learning toward physically consistent solutions, improving robustness and interpretability compared with purely data-driven FNO models. A key advantage of the proposed strategy lies in residual refinement, which mitigates the loss of high frequency and localized scattering information arising from truncated spectral representations in standard FNO. The proposed framework enhances the representation of localized scattering features and sharp spatial variations, leading to improved overall prediction fidelity for Lamb wave scattering fields. The benefits of this dual-stage learning strategy are clearly demonstrated by the results shown in

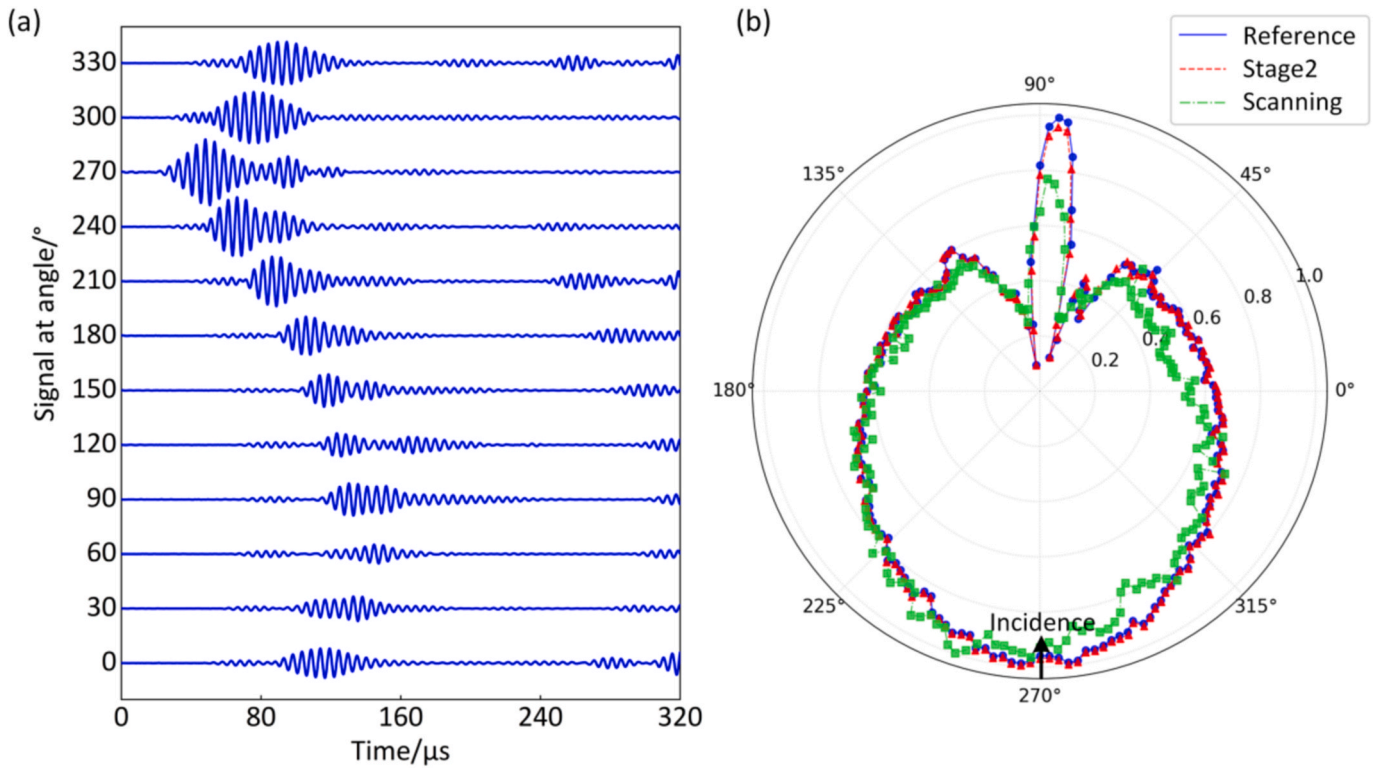


Fig. 13. Scanning signals and energy distribution pattern (the black arrow indicates the direction of incidence).

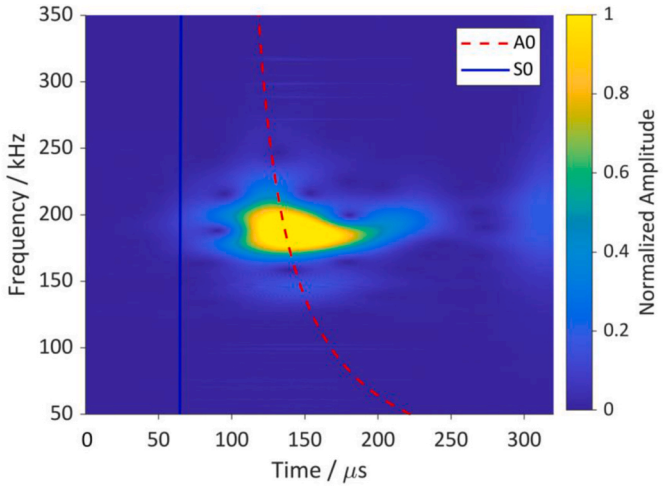


Fig. 14. CWT results of the scanning signal at 90°.

Figs. 8–10.

Another appealing feature of the framework is its conceptual extensibility. Forward scattering problems and the corresponding inverse problems are generally governed by the similar underlying physical equations. Therefore, the proposed dual-stage strategy can be naturally extended to inverse scattering and defect imaging tasks, where defect characteristics are inferred from measured wavefields. From this perspective, the present framework offers a common foundation for forward modeling and inverse analysis. In addition, beyond Lamb wave scattering in plates, the proposed method is inherently general and can be adapted to other guided wave problems by modifying the governing equations and input parameterization. This flexibility makes the approach a promising candidate for a wide range of guided wave NDT and SHM applications.

It is worth noting that, the proposed method is not intended to replace conventional numerical approaches such as finite difference or finite element methods, which remain highly reliable and flexible tools especially when the damage geometry is known in advance and only a limited number of forward simulations are required. Instead, the proposed framework serves as a complementary tool that offers advantages in scenarios demanding rapid, repeated forward predictions. Its primary value lies in enabling fast evaluation of multiple defect configurations, rather than outperforming traditional solvers in all use cases. After offline training, the proposed surrogate model facilitates rapid and repeated predictions, making it particularly attractive in the context of real-time or near-real-time wavefield computation. In such settings, surrogate approaches may become more competitive than repeated conventional simulations. This capability underscores the growing interest in learning-based alternatives in recent years and highlights the relevance of the proposed direction.

5.2. *Limitations and future perspectives*

Despite its advantages, the present study also has several limitations that warrant further investigation. First, the current implementation focuses on a single mode and employs a simplified governing equation for plate waves. This study mainly focuses on the out-of-plane displacement field, based on the Kirchhoff-Love plate theory, which is applicable to simulating the A0 mode and does not explicitly explain the transformation of A0-S0 mode. In fact, under the considered frequency-thickness conditions, the response is still dominated by the A0 mode, while the contribution of mode conversion S0 in the out-of-plane field is relatively limited. Additionally, the proposed dual-stage learning strategy is not restricted to the Kirchhoff-Love formulation itself. For more general scenarios involving multiple modes and mode conversions, it would be preferable to incorporate more comprehensive governing equations, such as the Navier elastodynamic equations. Extending the framework to the quantitative analysis of multimodal Lamb wave propagation and scattering, as well as to 3D wavefields, represents an

important direction for future work. Second, although the proposed framework benefits from physics embedding, it still relies on a relatively large dataset for training. Reducing data requirements remains a key challenge for practical deployment. Future research may draw inspiration from unsupervised or weakly supervised learning strategies, further exploiting physical constraints to improve performance under limited data conditions and enhance sample efficiency.

From an experimental perspective, the current validation employs machined defects with controlled geometries. While this enables experimental repeatability, such defects do not fully represent the complexity of real corrosion damage. More realistic defect generation methods, such as electrochemical corrosion, combined with high resolution thickness characterization using microscopy or profilometry techniques, would allow for more faithful experimental validation and further assess the robustness of the proposed method under practical conditions.

Addressing these limitations in a systematic manner will broaden the applicability of the proposed dual-stage strategy and further strengthen its relevance for real-world guided wave inspection problems.

6. Summary

This work presents a dual-stage FNO framework with physics embedded and residual refinement for solving Lamb wave scattering in plate structures. The embedding of governing equation improves the physical consistency of predicted wavefields and strengthens the coherence of the global scattering representation. Residual refinement further compensates for information loss arising in FNO learning, particularly in wavefield components sensitive to defects and high frequencies. Through this dual-stage design, the proposed framework achieves improved global wavefield prediction performance together with more accurate reconstruction of local scattering details, providing a more reliably description for Lamb wave scattering phenomenon. Numerical simulations demonstrate that the proposed framework improves wavefield prediction performance, with pronounced error reduction in regions where conventional FNO models typically lose accuracy. The prediction error has achieved an overall reduction, with a performance improvement of approximately 30% to 45%. The scanning experiment further shows that the predicted wavefield energy distributions closely match the reference solution and measurement results, effectively capturing the scattering behavior. These results provide good evidence for the effectiveness of the proposed dual-stage FNO strategy and establish it as a promising modeling paradigm for guided wave scattering problems. The proposed framework has the potential to be extended to more complex and realistic engineering scenarios, involving multimodal wave propagation and heterogeneous structures, providing a flexible foundation for future research in advanced guided wave NDT.

CRediT authorship contribution statement

Linfeng Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Hongyan Zhang:** Visualization, Validation, Software, Methodology, Conceptualization. **Zhao Wang:** Visualization, Software, Methodology, Investigation, Data curation. **Yizhe Gao:** Validation, Software, Resources, Conceptualization. **Jing Hou:** Resources, Project administration, Funding acquisition, Data curation. **Yang Liu:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

The out-of-plane displacement fields for both a pristine plate and a plate containing a single irregular defect were solved using a commercial finite element method (3D FEM) and the finite difference scheme based on Kirchhoff-Love plate equation adopted in this work (KL FDM). For both approaches, the simulation settings were kept fully consistent with those described in Section III.A, including the excitation location and frequency, computational domain, PMLs, spatial discretization, time step, and total simulation duration. The pristine plate had a uniform thickness of 3 mm, while the thickness distribution of the plate with a single irregular defect is shown in Fig. A.1.

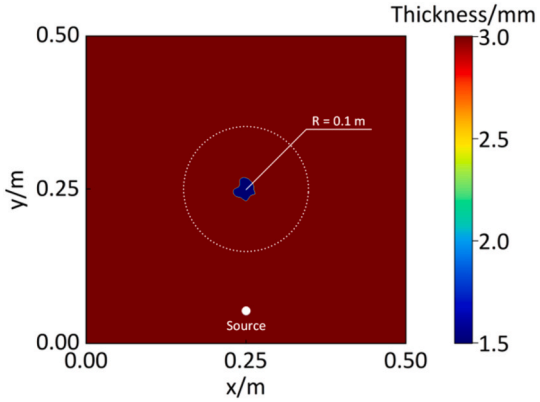


Fig. A.1. Thickness distribution of the irregular defect.

After obtaining the wavefields, time-domain signals were uniformly sampled at 360 angular positions along a circular trajectory centered at the computational domain, with a radius of 0.1 m. Each signal corresponds to a distinct angular position. The maximum amplitude of each signal was extracted, normalized, and mapped onto a polar coordinate system. For both numerical methods, the energy distribution patterns of the pristine plate,

the plate with the irregular defect, and the corresponding perturbation components obtained by subtracting the pristine signals from the defective ones were compared, as shown in Fig. A.2. The black arrow indicates the direction of incidence. The observed consistency between the two sets of results verifies the effectiveness of the finite difference scheme adopted in this work in accurately capturing Lamb wave propagation and scattering behavior.

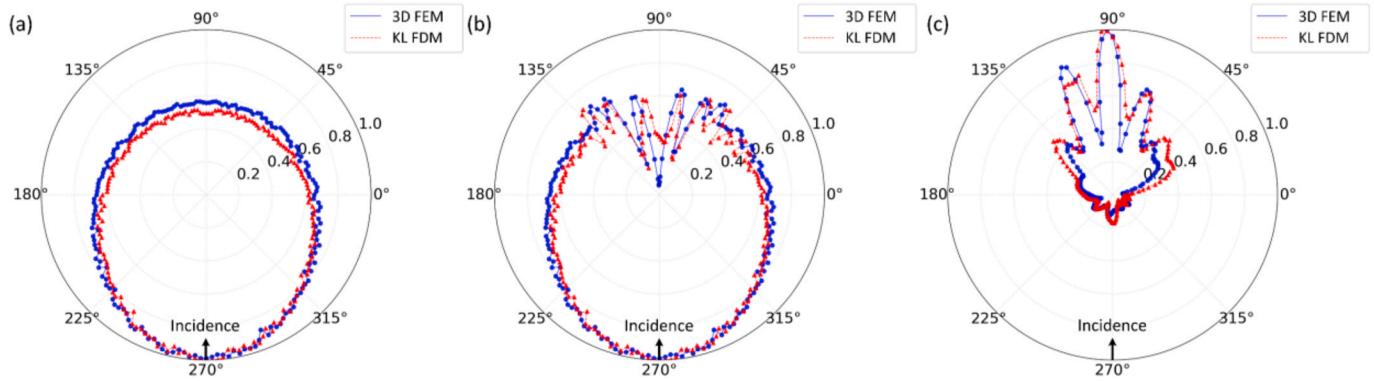


Fig. A.2. The energy distribution patterns of (a) pristine plate, (b) plate with the irregular defect, and (c) the corresponding perturbation components (the black arrow indicates the direction of incidence).

Data availability

Data will be made available on request.

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